

Linear Algebra — Exercises

Linear independence. Bases and dimension

Exercise 1. Decide whether the given sequence of vectors is linearly independent in the indicated vector space:

- (a) $((1, 0, 1), (1, 1, 0), (0, 1, 1))$ in $\mathbb{R}^3$ over $\mathbb{R}$
- (b) $((1, 0, 1), (1, 1, 0), (0, 1, 1))$ in $\mathbb{Z}_2^3$ over $\mathbb{Z}_2$

Exercise 2. Let V be any vector space over a field $\mathbb{K}$ and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n, \mathbf{u} \in V$. Decide whether the given sequence of vectors is linearly independent:

- (a) $(\mathbf{v}_1, \mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3, \dots, \mathbf{v}_1 + \dots + \mathbf{v}_{n-1} + \mathbf{v}_n)$ if the sequence $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ is linearly independent
- (b) $(\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_2 + \mathbf{v}_3, \dots, \mathbf{v}_{n-1} + \mathbf{v}_n, \mathbf{v}_n + \mathbf{v}_1)$ if the sequence $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ is linearly independent
- (c) $(\mathbf{v}_1 + \mathbf{u}, \mathbf{v}_2 + \mathbf{u}, \dots, \mathbf{v}_n + \mathbf{u})$ if the sequence $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ is linearly independent and $\mathbf{u} \in V$ is any vector
- (d) S , where S is a subsequence of a linearly independent sequence T of vectors from V
- (e) S , where S is a sequence of vectors from V containing a linearly independent subsequence T
- (f) $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$, where $\mathbf{v}_i \in \text{span}(S_i)$ and $\mathbf{v}_i \neq \mathbf{0}$ for $i = 1, 2, \dots, n$, the sets $S_i \subseteq V$ are finite and pairwise disjoint, and the set $S_1 \cup S_2 \cup \dots \cup S_n$ consists of linearly independent vectors.

Exercise 3. Find the dimension of each of the following vector spaces:

- (a) $\mathbb{C}$ over $\mathbb{R}$
- (b) $\mathbb{K}^n$ over any field $\mathbb{K}$
- (c) $\text{span}(\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{n-1}\})$, where $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ is a basis of a vector space V over a field $\mathbb{K}$
- (d) $\text{span}(\{\mathbf{v}_n, \mathbf{v}_{n-1}, \dots, \mathbf{v}_1\})$, where $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ is a basis of a vector space V over a field $\mathbb{K}$
- (e) $\{(x, y, z, t) \in \mathbb{R}^4 \mid x + y + z + t = 0\}$
- (f) $\{(x, y, z, t) \in \mathbb{R}^4 \mid 2x + y = z - t\}$
- (g) $\text{span}(\{(1, 2, 1, 0), (1, 3, 1, 1), (0, 1, 2, -1), (-3, 1, 2, 2)\})$ over $\mathbb{R}$
- (h) $2^{\{a,b,c\}}$ over $\mathbb{Z}_2$ (with the symmetric difference as vector addition).

Exercise 4. Calculate the coordinates of the vector $\mathbf{v}$ with respect to the basis S :

- (a) $\mathbf{v} = (1, 2, 3, 4)$, $S = ((1, 1, 1, 1), (1, 1, 1, 0), (1, 1, 0, 1), (1, 0, 1, 1))$
- (b) $\mathbf{v} = \{a, b\}$, $S = (\{a\}, \{b\}, \{c\})$ in the vector space $2^{\{a,b,c\}}$ over $\mathbb{Z}_2$
- (c) $\mathbf{v} = \mathbf{v}_1$, $S = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$
- (d) $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2 + \dots + \mathbf{v}_n$, $S = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$.

Exercise 5. Prove that if $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ is a basis of a vector space V , then so is $(\mathbf{v}_1, \mathbf{v}_1 + \mathbf{v}_2, \dots, \mathbf{v}_1 + \mathbf{v}_2 + \dots + \mathbf{v}_n)$.

Exercise 6. Prove that $\dim \text{span}(A \cup B) \leq \dim \text{span}(A) + \dim \text{span}(B)$ for subsets A, B of a vector space V .